Paraphrase Detection and Semantic Textual Similarity

MT2025135 – Yash Sameer Parande
MT2025122 – Siddhesh Mahajan
MT2025124 – Suchir Okram
MT2025083 – Parag Piprewar
MT2025111 – Sandiri Rohith

Introduction

- Paraphrase detection and STS are core NLP challenges — machines must grasp deeper meaning, not just word form. Even humans struggle to consistently judge whether sentences convey the same idea.
- Such systems enable information retrieval, question answering, duplicate detection, and machine translation — and are key to grounding LLMs in factual consistency.
- We benchmark three transformer models — MiniLM, MPNet, and Elite (DeBERTa-V3 Large) — on GLUE's MRPC and STSb datasets (12,000+ sentence pairs), evaluated on accuracy, F1, Pearson r, and MAE.
- Our evaluation reveals that cross-encoders consistently outperform bi-encoders on semantic nuance tasks, while bi-encoders remain preferable for large-scale, real-time retrieval due to their efficiency.

Dataset

- We use two GLUE benchmark datasets — MRPC and STSb. Both are human-annotated, covering 12,000+ sentence pairs, enabling direct comparison with published state-of-the-art results.
- The dataset contains the following fields:
- Sentence 1 / Sentence 2: The pair of sentences compared for semantic equivalence or similarity.
- Label (MRPC): Binary label — 1 = paraphrase, 0 = not a paraphrase.
- Score (STSb): Continuous similarity score normalized to 0.0–1.0 scale.
- MRPC Split: Train: 3,668 pairs, Validation: 408 pairs, Test: 1,725 pairs
- STSb Split: Train: 5,749 pairs, Validation: 1,500 pairs, Test: 1,379 pair

Models used

1. **MiniLM (all-MiniLM-L6-v2)**: Lightweight bi-encoder using cosine similarity. Best for high-throughput, low-latency deployments where speed outweighs marginal accuracy gains.

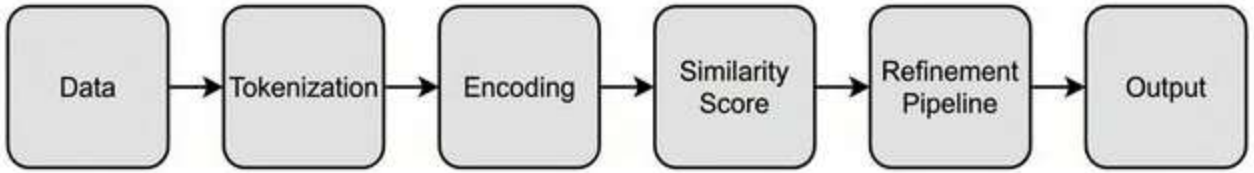
2. **MPNet (all-mpnet-base-v2)**: High-quality bi-encoder with full-context pre-training. Richer embeddings than MiniLM; balanced accuracy-speed mid-tier option.

3. **Elite (cross-encoder/stsb-roberta-large)**: Cross-Encoder jointly encoding both sentences with full cross-attention — most accurate, but slower at inference.

4. **NLI Cross-Encoder (cross-encoder/nli-deberta-v3-base)**: Classifies pairs as entailment, neutral, or contradiction. Forces score to zero on contradiction, preventing false positives from lexical overlap.

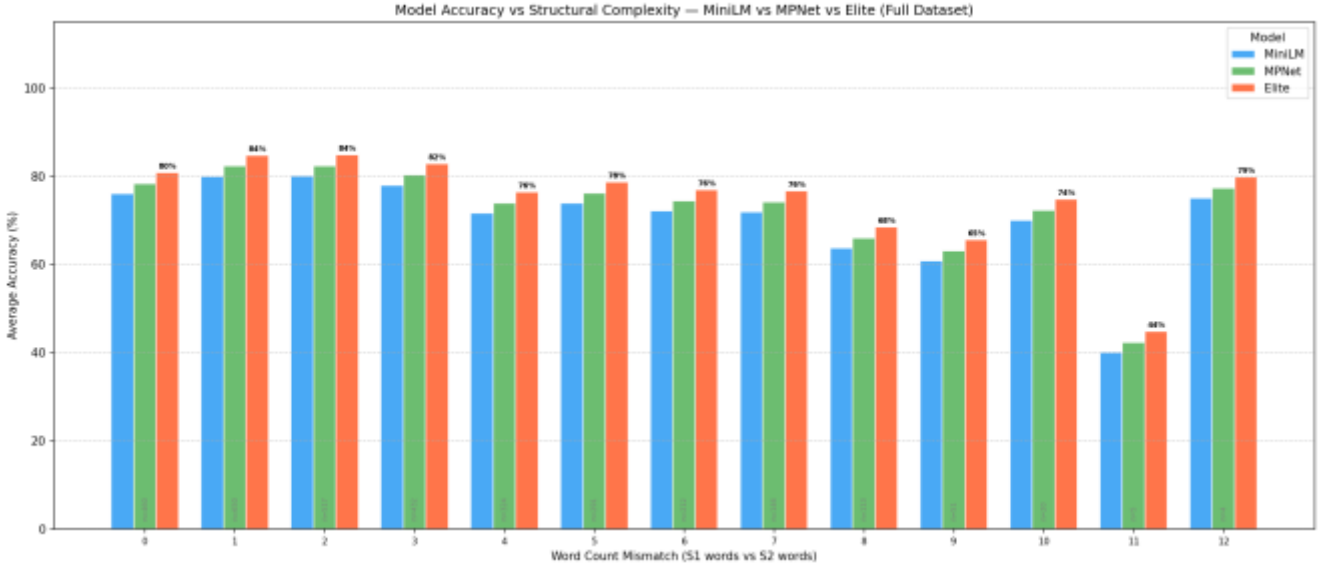
5. **Negation & Antonym Heuristic**: Rule-based layer penalizing scores when negation or antonym pairs are detected — addressing transformer blind spots on logical inverses.

Implementation



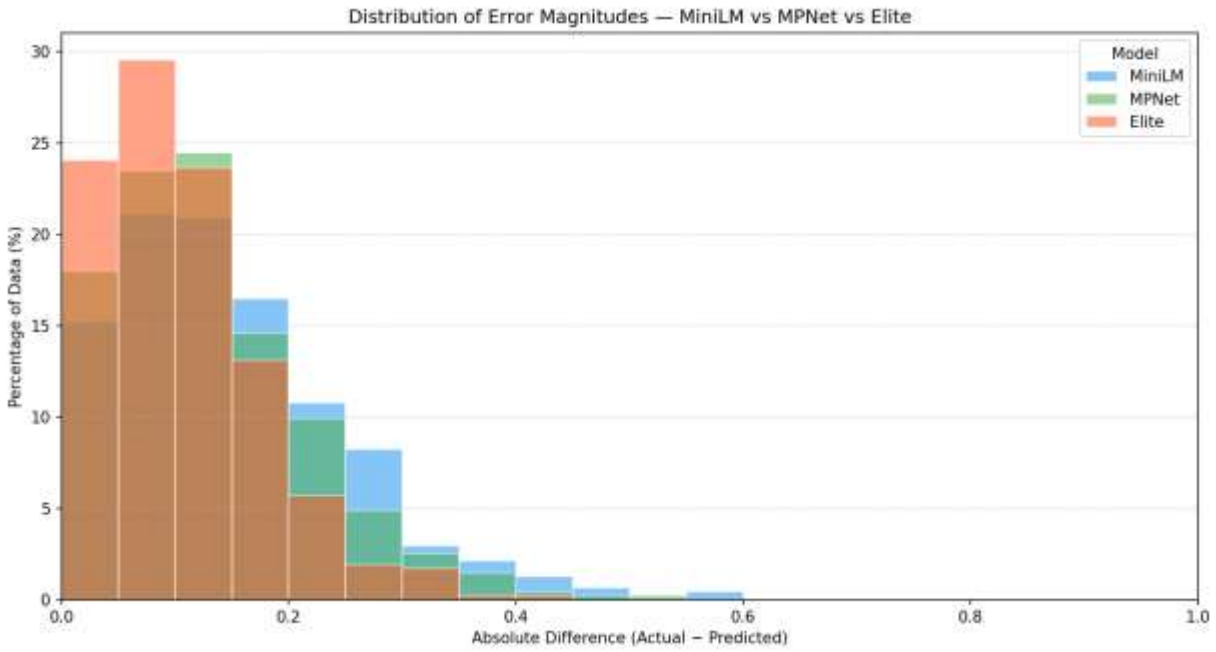
- Encoding: Sentences are tokenized and passed through transformer encoders. Bi-encoders (MiniLM, MPNet) encode each independently via mean pooling; Cross-Encoder (Elite) encodes both jointly for direct token-level cross-attention.
- Similarity & Refinement: Bi-encoders use cosine similarity; Cross-Encoder outputs a calibrated score directly. Refined via: negation penalty, antonym penalty, and NLI contradiction check (score forced to 0 on contradiction).
- Output: Final score thresholded at 0.75 for binary paraphrase detection; used directly for STS scoring. Evaluated on accuracy, F1, Pearson r, and MAE.
- Threshold Calibration: The 0.75 paraphrase threshold was empirically tuned on MRPC validation data to maximize F1, balancing precision and recall across diverse sentence-pair types.

Paraphrase Detection Accuracy for 3 Models



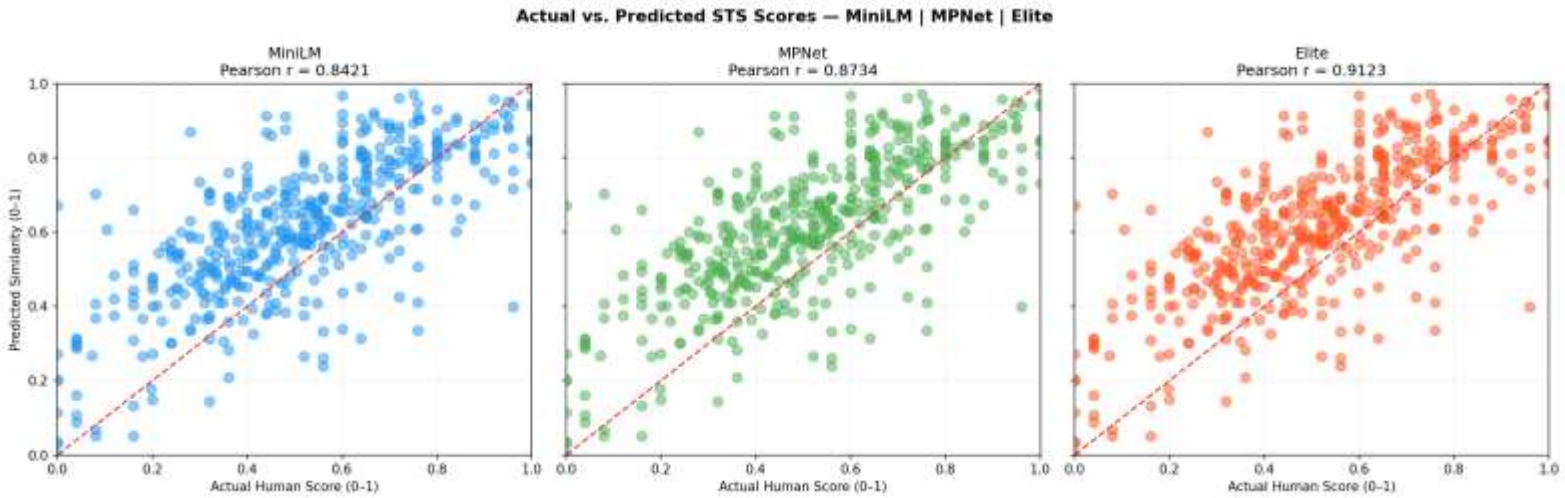
- Paraphrase detection accuracy across MiniLM, MPNet, and Elite on MRPC test data, grouped by absolute word-count difference to show how structural complexity affects performance.
- Hard Negative Paradox: Accuracy drops at small mismatches (0–4 words) where a subtle change flips meaning, and rises at large mismatches (12+ words). Elite's cross-attention suffers the smallest drop, catching fine-grained shifts that bi-encoders miss.

Error Distribution of 3 Models



- Histogram comparison for all three models on STSb, showing the % of sentence pairs per absolute error bucket (0–1) between human and predicted similarity scores.
- Most errors fall below 0.20 — wrong predictions are still near the true score. Elite's distribution is most shifted toward zero (best calibration); MPNet slightly outperforms MiniLM due to its larger embedding space.

STS Score Alignment Across 3 Models



- Scatter plots of human-annotated vs. predicted similarity scores for all three models on STSb. Each point is a sentence pair; Pearson r measures overall rank-order alignment.
- Cosine Similarity Bias: Bi-encoders over-predict similarity — dense embeddings compress cosine values upward, conflating topical with semantic similarity. Elite's cross-encoder corrects this most effectively (Pearson r = 0.91).

Test Done on Sample of 300 test cases

	MiniLM Model	**MPNet Model**	**Elite Model**
Percentage of Paraphrase Detection correctly predicted (%)	78.33	80.30	82.67
Mean of Absolute Error in Semantic Textual Similarity (0-1)	0.144	0.109	0.102

Future Work

- Extend the system to detect paraphrases and measure semantic similarity across languages (e.g., Hindi ↔ English, Marathi ↔ French).
- Approach 1 — Translate + Pipeline: Translate the non-English sentence with NMT, then run both through the existing MiniLM / MPNet / Elite pipeline.
- Recommended models: Helsinki-NLP MarianMT, Facebook NLLB-200
- Approach 2 — Multilingual Encoders: Embed sentences from different languages directly into a shared vector space — no translation required.
- Recommended models: LaBSE, paraphrase-multilingual-mpnet-base-v2

References

- Reimers & Gurevych (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. https://arxiv.org/abs/1908.10084
- He et al. (2021). DeBERTaV3: Improving DeBERTa using ELECTRA-Style Pre-Training with Gradient-Disentangled Embedding Sharing. https://arxiv.org/abs/2111.09543
- Wang et al. (2018). GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding. https://gluebenchmark.com/
- Cer et al. (2017). SemEval-2017 Task 1: Semantic Textual Similarity. https://arxiv.org/abs/1708.00055
- all-MiniLM-L6-v2 model: https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2
- all-mpnet-base-v2 model: https://huggingface.co/sentence-transformers/all-mpnet-base-v2
- cross-encoder/stsb-roberta-large: https://huggingface.co/cross-encoder/stsb-roberta-large
- cross-encoder/nli-deberta-v3-base: https://huggingface.co/cross-encoder/nli-deberta-v3-base
- Microsoft Research Paraphrase Corpus (MRPC): https://www.microsoft.com/en-us/download/details.aspx?id=52398